PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT AUTHORITY
KINONDONI MUNICIPAL COUNCIL
FORM FOUR DISTRICT PRE-MOCK EXAMINATION

CODE: 041/1 **BASCI MATHEMATICS**

Time: 3 Hours **Year: March 2024**

Instructions

1. This paper consists of sections A and B with a total of 14 questions.
2. Answer **all** questions
3. Each question in section A carries six marks while each questions in section B carries 10 marks
4. All necessary working and answer for each question must be shown clearly.

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SECTION A

1. (a) If a, b, c are three digits, abc and cba are two numbers. When abc + cba = 444, b = 2. Find the value of a + b + c

   (b) Mr. Mosha used tiles to build his floor of his room measuring 18cm by 24cm. If he used only whole ones and they were a like.

   (i) What was the greatest size of each tiles?

   (ii) How many pieces of tiles used to cover the room?

2. (a) I f $4^{x+2} + 4^{x+5} = 65$ Find the value of X

   (b) Evaluate the following without using mathematical table

   $\frac{1}{2}\log 64 + \log 50 - \log 4$

3. (a) Given set A = $\{x: -5 \leq x \leq 2\}$ and B = $\{x: 1 \leq x \leq 4\}$. List all subsets of A∩B

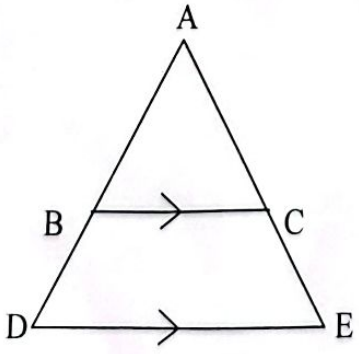
   (b) A box contains 6 red toys and 4 blue toys. Two toys are picked from the box at random one after another without replacement. Draw a tree diagram to illustrate the probability of these toys and hence determine the probability that the toys picked up are of different colours.

4. (a) Find the coordinates of the point on the y-axis which is equidistant from the point (8, 4) and (6, 6)

   (b) The straight line through the point C (1, -2) and D (3, 4) meets y-axis at point E. Find the coordinates of point E.

5. (a) Two in angles are similar. The ratio of their areas is 64 : 49. What is the ratio of their corresponding length.

   (b) If . Δ ABC and Δ ADE are similar. If AB = 8cm, AC = 9cm, BC = 6cm and AD = 12cm. Calculate the length AE and DE



6. (a) A piece of length 7.42m is cut off from a string that is 13.5m long. If the remaining part of the string is divided into equal pieces of length 32cm, how many pieces are there?

   (b) If y varies inversely as $\sqrt{x}$ and x is multiplied by n. What is the ratio of the first y and the second y?

7. (a) The volume of two tanks are in the ratio 2 : 5. If the volume of the first tank is 2600 litres. Find the total volume of the two tanks.

   (b) Three quarters of the exterior angle of a regular polygon is half of its interior angle. Find the number of sides of this polygon.

8. (a) The sum of the first two terms of a certain GP is 10, while the sum of the first 4 terms is 4 times the sum of the first 2 terms. Find the first term and the common ratio of this GP.

   (b) Given that 49, y and 81 are consecutive terms of certain G. Find

   (i) The value of y

   (ii) The geometric mean

9. (a) Given $\sin 3x = \cos 2x$. Find the value of x when $0^\circ < x < 90^\circ$

   (b) Given $\sin Q = 0.8$ and Q is an obluse angle. Find: (i) $\cos Q$ (ii) $\tan Q$

10. (a) Solve the equation $2x^2 - 3x - 5 = 0$ by using completing the square method.

    (b) A father is 3 times as old as her child, seven years ago the product o their ages was 104. Find the present age of the father.

SECTION B

11. (a) The height in cm of 100 people was recorded as shown in the table below;

Height	160	165	170	175	180	185
Frequency	2	12	32	24	x	8

   (i) Find the value of x

   (ii) Evaluate the median height.

   (b) Prove that if one side of a cyclic quadrilateral is produced, the exterior angle so formed is equal to the opposite interior angle.

12. (a) VABCD is a pyramid with VA = VB = VD = 5cm and ABCD is a square base of side 4cm each. Assume that the centre of the base is at point V. Find

   (i) The angle between VA and base ABCD

   (ii) The volume of the pyramid.

   (b) A ship sails from point A(40° S, 30° W) due west along the same latitude to point B for 1000 km. Find the latitude and longitude of point B (leave your answer in nearest degree).

13. (a) Find the image of the point (3, 2) that reflected in the line $y = -x$ followed by $y = x$

(b) A translation takes every point a distance 1 unit to the left and two units downwards on the xy plane. Find where it takes (3, 2).

(c) Find the image of the point (1, 2) under a rotation through 180^0 anticlockwise followed by 90^0.

14. (a) Given $f(x) = x^2 - 2x - 3$. By using completing the square method, Find;

(i) Turning point

(ii) Domain and range

(iii) The line of symetry

(iv) The maximum or minimum value

(b) Given $f(x) = \frac{3}{4x-2}$ and $f^1(x) = \frac{ax+3}{4x}$

Find the value of a.